

4 (**short**) A fluid of density ρ is confined in a reservoir by a gate of length L under the effect of gravity g , as shown in Fig. 2. Both the reservoir and the gate are homogeneous in the direction into the page, and the gate is held by a torsional spring. The restoring moment per unit length out of the page from the spring is $M = K\theta$, where K is the spring constant and θ the angle of the gate with the vertical.

- (a) Find the moment M produced by the hydrostatic pressure on the gate hinge as a function of ρ , g , h and θ . [5]
- (b) Find the value of K in terms of ρ , g and L for which the fluid will spill for $h = \sqrt{3}L/2$. [5]

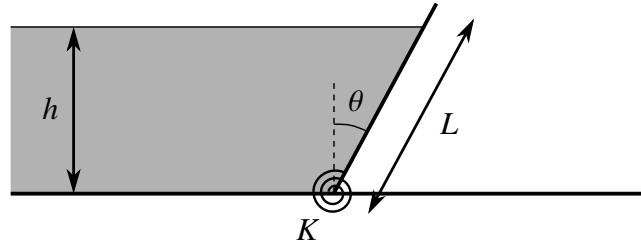


Fig. 2